



Clinical RoPE (cRoPE) score predicts patent foramen ovale detection among stroke patients: a multicenter observational study

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Abstract

Background The role of patent foramen ovale (PFO) in cryptogenic stroke (CS) is debated. Tools to predict PFO occurrence and attributable fraction are needed to guide cost-effective diagnostics and treatment. Risk of Paradoxical Embolism (RoPE) score relies on neuroimaging findings, which might be inconclusive in up to 30% of cases.

Methods We developed a clinical-based easy tool to predict the presence and attributable fraction of PFO in CS patients, without using neuroimaging. The clinical RoPE (cRoPE) score, ranging 1–10, was elaborated through Delphi method from the original RoPE score, replacing cortical infarction with the Oxfordshire Community Stroke Project (OCSP) classification (lacunar stroke = 0 points, other subtypes = 1 point). Then, from the SISIFO (Studio Italiano di prevalenza nello Stroke Ischemico di pervietà del Forame Ovale, or Prevalence of Patent Foramen Ovale in Ischemic Stroke in Italy) study, a multicenter, prospective study on consecutive acute ischemic stroke patients ($n = 1130$) classified by Trial of Org 10172 in Acute Stroke Treatment (TOAST) and OCSP criteria and undergoing PFO testing, we selected the VV-CDC cohort (Vibo Valentia, Città di Castello, $n = 323$) to test the accuracy of cRoPE in predicting PFO detection. We compared cRoPE with RoPE to verify cRoPE reliability. Finally, we tested, through ROC analysis, the performance of cRoPE depending on TOAST classification.

Results Overall, PFO was detected in 21% in VV-CDC and in 23.4% in remaining SISIFO cohort ($n = 807$). $cRoPE_{AUC}$ and $RoPE_{AUC}$ were similar in VV-CDC. cRoPE performance was comparable with RoPE among CS ($cRoPE_{AUC}$ 0.76, 95%CI 0.67–0.85, $RoPE_{AUC}$ 0.75, 95%CI 0.66–0.84). Moving to the remaining SISIFO cohort, cRoPE confirmed satisfactory accuracy in predicting PFO detection in CS patients ($cRoPE_{AUC}$ 0.71, 95%CI 0.66–0.78, $p = 0.032$).

Conclusions Conclusions: cRoPE might help in stratification of patients with CS, allowing accurate esteem of the likelihood of PFO to be found, especially in cases when neuroimaging is inconclusive.

Keywords Stroke · Paradoxical embolism · Patent foramen ovale · Cryptogenic stroke

David Giannandrea and Michele Romoli contributed equally to this work.

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Introduction

Despite extensive diagnostics in up to 25% of patients with stroke, no specific cause can be identified, leading to the diagnosis of cryptogenic stroke (CS) [1]. Among patients with CS, patent foramen ovale (PFO) has considerable prevalence [2, 3]. The Risk of Paradoxical Embolism (RoPE) score defines the likelihood of PFO being the cause of stroke, allowing to grossly differentiate between incidental and causal PFO among patients with CS [4]. RoPE score reflects the absence of vascular risk factors (including hypertension, diabetes, previous stroke or TIA, and smoke) and the presence of specific demographic (age groups) and neuroradiological (cortical infarct on imaging) characteristics. The higher the RoPE score, the more likely the stroke is due to PFO [4].

However, the accuracy of the scoring system is still debated, in particular due to the lack of inclusion of stroke severity data and anatomical PFO characteristics, as well as for controversies on neuroimaging [4, 5]. Indeed, beyond availability issues, brain imaging fails to demonstrate an ischemic lesion in up to 21% of CS [6, 7], leaving the “cortical infarct” item rating uncertain. In addition, debate still exist on the relationship between cortical infarction in patients with PFO [8–12], even more after revascularization treatments came to light, substantially impacting on brain MRI findings, and possibly misleading the rating of the neuroimaging item [13]. Hence, the need for a purely clinical score, applicable at stroke onset and without the compelling need of neuroimaging data, might help clinicians in defining the role of PFO as a possible causal agent among patients with stroke.

Using data from the SISIFO study, a multicenter, observational, prospective study on consecutive patients with acute ischemic stroke with systematic screening for PFO [2], we aimed to validate the cRoPE score, with clinical and demographic features only, to allow clinicians to estimate the likelihood of finding PFO depending on stroke subtype.

Methods

Study population

The SISIFO study is a multicenter, prospective, cross-sectional observational study on consecutive patients admitted to 13 Italian centers for acute ischemic stroke [2]. The study protocol was approved by the Ethical Committee of Coordinator Centre (4.2011.22) and by each Local Ethical Committees subsequently. Methods have been already reported [2]. Briefly, consecutive patients with ischemic stroke admitted to 13 Italian stroke units were enrolled giving informed consent between November 2011 and February 2013. For all patients,

demographic, clinical, and neuroradiological data were collected. Ischemic strokes were classified by clinical presentation and etiology, using the Oxfordshire Community Stroke Project (OCSP) [2] and the Trial of Org 10172 in Acute Stroke Treatment (TOAST) classification [2, 14]. During the diagnostic evaluation, all patients underwent the best available brain imaging modality (CT or MRI), according to on sites protocols for acute ischemic stroke, electrocardiogram, standard laboratory blood tests, Doppler ultrasound study of extracranial arteries, transthoracic echocardiography (TTE), and transcranial Doppler (TCD) or transcranial color-coded Doppler (TCCD) [2]. Further diagnostic was provided if uncommon causes of ischemic stroke were raised.

Definitions, original RoPE, and clinical ROPE (cRoPE)

Among the 1130 patients prospectively enrolled in the SISIFO study [2], data from two centers (Vibo Valentia, VV; Città di Castello, CDC; together with VV-CDC cohort, $n = 323$) were extracted to develop the cRoPE model. Two neurologists analyzed baseline characteristics in the derivation cohort at admission, as well as stroke diagnosis, OCSP, and TOAST at discharge [2, 15].

For the development of the cRoPE score, we applied the Delphi method [16] among the study group to define the clinical features to replace the item “cortical infarcts on imaging” of the original RoPE score [4]. Unanimous agreement was reached to (i) provide a binary score in order to maintain the cRoPE score range equal to the original RoPE score, (ii) replace the neurological item with a clinical data easy to retrieve, and (iii) to implement a clinical variable reflecting lesion localization. Drafts from authors were compared throughout four rounds, until unanimous agreement was reached. Among all scales considered, the OCSP classification was chosen, being already included in the original SISIFO study, and given its reliability, easiness of use, rapidity, and its radiological correlates [2, 4, 12, 17–19]. Then, the item “cortical infarct” was replaced with OCSP classification, to be scored as follows: “0” for lacunar syndrome (LACS), “1” for total anterior (TACS), partial anterior (PACS), and posterior syndromes (POCS) [2]. The cRoPE score was defined as follows: no history of hypertension (1 point), no history of diabetes (1 point), no history of stroke or TIA (1 point), non-smoker (1 point), OCSP TACS/PACS/POCS (1 point), age 18–29 (5 points), 30–39 (4 points), 40–49 (3 points), 50–59 (2 points), 60–69 (1 point), and 70-year old or more (0 points). cRoPE ranged from 0 to 10, as the original RoPE [4].

OCSP classification was attributed after consensus, taking into account the initial clinical evaluation and follow-up evaluation after 48 h. Only clinical syndromes before treatment were considered.

We compared the cRoPE model with the original RoPE score in the VV-CDC cohort. We then tested cRoPE predictive value for PFO detection among the entire SISIFO population. Furthermore, to define the predictive value of cRoPE among patients with stroke subtypes, we identified patients with CS according to TOAST criteria and patients with embolic stroke of undetermined source (ESUS) [20] within the extracted VV-CDC cohort. We then applied both original RoPE and cRoPE scores among patients with CS and ESUS, to verify their performance in terms of prediction of PFO. We then validated cRoPE among all patients with ESUS in the SISIFO cohort. Attributable fraction was calculated for CS patients with Bayes' theorem and estimate of PFO in general population fixed to 25%, according to previously reported paradigm [4], with cRoPE score segmented in low (0–5), mid (6–7), and high (8–10). Data sharing requests will be evaluated by the corresponding author.

Statistical analysis

Data are expressed as mean $\pm$ SD or proportions with 95% CIs as appropriate. Overall, 1130 patients were retrieved from the SISIFO study [2]. To assess the association between PFO and potential risk factors, OR and 95% CIs for categorical variables, t-test for continuous variables (e.g., age), and non-

parametric tests for ordinal variables were used. Reliability analysis for cRoPE score was performed on selected items, while correlation of cRoPE scores with original RoPE score was tested using Pearson's correlation. Receiver operating characteristics (ROC) analysis was used to verify the performance of cRoPE in predicting PFO detection among patients in the VV-CDC cohort, comparing it with the original RoPE score. We then proceeded to compare, in VV-CDC cohort, RoPE and cRoPE performance in detecting PFO within patients with CS and ESUS. Calibration was examined by dividing patients in quintiles according to their predicted risk, and the mean predicted risk per quintile group was subsequently plotted against the observed risk per quintile group. Calibration accuracy was evaluated through the calibration slope and plot. Finally, we extended the evaluation of cRoPE to the remaining SISIFO cohort, testing its accuracy depending on stroke subtype. Original RoPE and cRoPE scores were applied to the validation cohort to calculate the risk of PFO in all patients. Such analysis was replicated for patients with CS according to TOAST criteria and for patients with ESUS. DeLong's test was used to compare TOAST-specific cRoPE AUC in the SISIFO cohort (excluding VV-CDC). Statistical significance was set at $p < 0.05$ two-tailed. All analyses were performed with SPSS 25.0 and R v. 3.5.

Table 1 Cohort characteristics

	VV-CDC	Other sites	Overall SISIFO [2]	<i>p</i> value
<i>n</i>	323	807	1130	0.01
Age	72.2 $\pm$ 13.8	66.6 $\pm$ 14.4	68.2 $\pm$ 14.5	0.019
Gender (female)	147 (45.5%)	306 (37.9%)	453 (40.1%)	0.04
NIHSS	5.5 $\pm$ 4.7	6.7 $\pm$ 5.6	6.3 $\pm$ 4.0	0.01
TOAST				
Large artery atherosclerosis	47 (14.6%)	162 (20.2%)	209 (18.6%)	0.04
Cardioembolic	69 (21.4%)	135 (16.8%)	204 (18.1%)	
Small vessel disease	44 (13.6%)	137 (17.1%)	181 (16.1%)	
Other determined etiology	0 (0%)	12 (1.5%)*	12 (1.1%)	
Undetermined	163 (50.5%)	356 (44.4%)	519 (46.1%)	
OCSF				
LACS	94 (29.1%)	206 (25.5%)	300 (26.5%)	ns
PACS	151 (46.7%)	359 (44.5%)	510 (45.1%)	
POCS	58 (18%)	162 (20.1%)	220 (19.5%)	
TACS	20 (6.2%)	72 (8.9%)	92 (8.1%)	
Unclassified	0 (0%)	8 (1%)	8 (0.7%)	
Risk factors				
Hypertension	253 (78.3%)	540 (66.9%)	793 (70.2%)	0.001
Smoke	53 (16.4%)	232 (29.7%)	285 (25.8%)	0.001
Atrial fibrillation	81 (25.1%)	171 (21.2%)	252 (22.3%)	ns
PFO	68 (21.1%)	189 (23.4%)	257 (22.7%)	ns

VV-CDC, Vibo Valentia, Città di Castello cohort; LACS, lacunar stroke; NIHSS, National Institutes of Health Stroke Scale; OCSF, Oxfordshire Community Stroke Project; PACS, partial anterior circulation stroke; PFO, patent foramen ovale; POCS, posterior circulation stroke; TACS, total anterior circulation stroke; TOAST, Trial of Org 10,172 in Acute Stroke Treatment

Results

Cohort characteristics

Among the 1130 patients included in the SISIFO study [2], 323 were retrieved from the two centers promoting this study (CDC and VV). Demographic and clinical data on study cohort are reported in Table 1. Mean age was 72.2 years in VV-CDC cohort, slightly older than the overall SISIFO (68.2, $p = 0.019$), and with female gender more represented (45.5% vs 40.1%, $p = 0.04$). No differences in OCSP distribution in VV-CDC and whole cohort were detected. Patients in VV-CDC were more frequently hypertensive (78.3% vs 70.2%) and had less smoking habit (16.4% vs 25.8%).

VV-CDC cohort and cRoPE validation

In VV-CDC cohort, PFO was detected in 21.8% of cases ($n = 68$) (Table 2). Patients with PFO were younger (66.6 vs 73.6, $p = 0.001$), more gender-balanced (female 45.5% vs 37.9%, $p = 0.04$), and less frequently diabetic (13.2% vs 25.9%, $p = 0.02$) compared with those without PFO. No differences in NIHSS, distribution of TOAST, and OCSP classification subtypes were detected. Mean RoPE score was significantly higher in PFO group (4 vs 3.4, $p = 0.002$), as was for cRoPE score (4.3 vs 3.7, $p = 0.003$). cRoPE had optimal reliability (Chronbach's $\alpha = 0.89$). Correlation analysis demonstrated

that cRoPE score and original RoPE score strictly correlated (Pearson correlation 0.914, $p < 0.01$).

cRoPE and RoPE comparison

In the VV-CDC cohort, the accuracy of cRoPE did not differ from RoPE in predicting PFO (cRoPE_{AUC} 0.62, 95%CI 0.54–0.70 vs RoPE_{AUC} 0.63, 95%CI 0.55–0.70, $p = 0.82$). The performance of cRoPE was similar to RoPE score among all TOAST subgroups, including CS (cRoPE_{AUC} 0.76, 95%CI 0.67–0.85 vs RoPE_{AUC} 0.75, 95%CI 0.66–0.84, $p = 0.9$) (Fig. 1a). Among patients with diagnosis of embolic stroke of undetermined etiology (ESUS, $n = 98$), 22 (22.4%) had PFO. In ESUS, cRoPE had similar accuracy to original RoPE in predicting PFO detection (cRoPE_{AUC} 0.77, 95%CI 0.66–0.82 vs RoPE_{AUC} 0.73, 95%CI 0.61–0.84, $p = 0.8$) (Fig. 1b). The calibration pattern of cRoPE was satisfactory according to the visual inspection of the calibration plot (supplementary material). A cRoPE score > 5 was 89% specific and 72.7% sensitive in predicting the presence of PFO among ESUS patients.

cRoPE identifies CS patients with PFO

Accuracy of cRoPE was replicated in the remaining SISIFO cohort (cRoPE_{AUC} 0.61, 95%CI 0.56–0.66). Refining ROC analysis depending on TOAST classification, cRoPE had

Table 2 Demographic and clinical features in patients in Vibo Valentia and Città di Castello (VV-CDC) cohort

<i>n</i>	PFO 68	No PFO 255	<i>p</i> value
Age	66.6 ± 14.8	73.6 ± 13.3	0.001
Gender (female)	147 (45.5%)	306 (37.9%)	0.04
NIHSS	5.5 ± 4.0	5.5 ± 4.9	Ns
TOAST			
Large artery atherosclerosis	5 (7.4%)	42 (16.5%)	Ns
Cardioembolic	18 (26.5%)	51 (20%)	
Small vessel disease	12 (17.6%)	32 (12.5%)	
Other determined etiology	0 (0%)	0 (0%)	
Undetermined	33 (48.5%)	130 (51%)	
OCSP			
LACS	20 (29.4%)	74 (29%)	Ns
PACS	32 (47.1%)	119 (46.7%)	
POCS	14 (20.6%)	44 (17.3%)	
TACS	2 (2.9%)	18 (7.1%)	
Risk factors			
Diabetes	9 (13.2%)	66 (25.9%)*	0.02
Hypertension	51 (75%)	202 (79.2%)	Ns
Smoke	13 (19.1%)	40 (15.7%)	Ns
RoPE score	4.0 ± 1.5	3.4 ± 1.2	0.002
Clinical RoPE score (cRoPE)	4.3 ± 1.5	3.7 ± 1.2	0.003

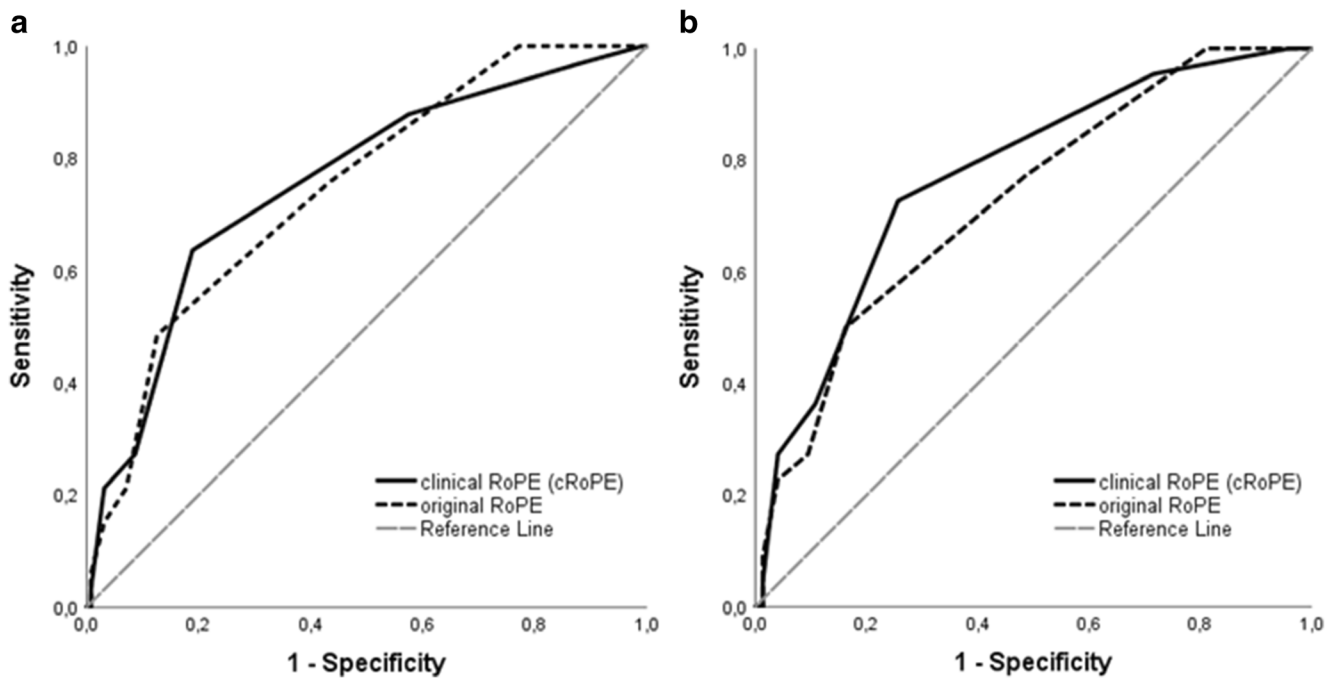


Fig. 1 ROC analysis among patients with cryptogenic stroke (a) and embolic stroke of undetermined source (b)

unsatisfactory performance in predicting PFO among patients with large artery atherosclerosis, small vessel disease, and other determined etiology (cRoPE_{AUC}, respectively, 0.43, 0.51, and 0.44), and limited predictive value in cardioembolic stroke (cRoPE_{AUC} 0.63, 95%CI 0.48–0.78, $p = 0.07$). However, satisfactory performance was found among patients with undetermined etiology (cRoPE_{AUC} 0.71, 95%CI 0.66–0.78), with cRoPE being significantly more accurate among

the latter compared with other TOAST subtypes ($p = 0.05$). In the latter group, cRoPE scores > 5 were 46% sensitive and 88% specific for PFO detection (scores > 3 had 80% sensitivity and 50% specificity) (Fig. 2a). Pooling together the whole SISIFO cohort, cRoPE had satisfactory accuracy in predicting PFO (cRoPE_{AUC} 0.72, 95%CI 0.67–0.77, $p = 0.02$) (Fig. 2b) and was significantly more accurate in CS compared with other TOAST stroke subtypes ($p < 0.01$).

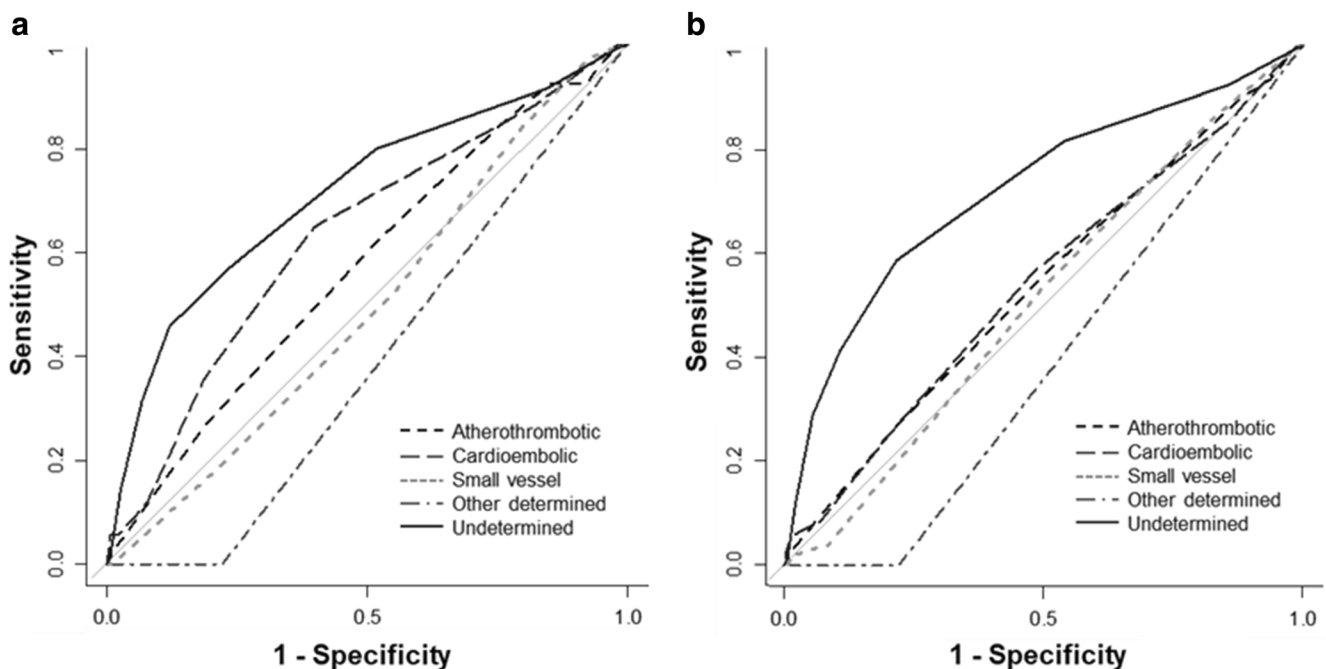


Fig. 2 ROC analysis among patients with cryptogenic stroke in the cohort not involved in validation ($n=807$) and in the whole SISIFO cohort ($n=1130$)

Table 3 PFO prevalence and attributable fraction among cryptogenic stroke patients

cRoPE score	n	PFO	PFO prevalence, % (95% CI)	PFO attributable fraction, % (95% CI)
≤ 5	421	81	19 (15–23)	0 (0–21)
6–7	74	41	55 (44–67)	73 (58–83)
≥ 8	24	16	67 (48–86)	83 (64–94)

PFO attributable fraction in CS patients

The attributable fraction, estimated using Bayes' theorem and a control rate of 25%, ranged from 0% (95%CI 0–21) for patients with cRoPE score ≤ 5 to 83% (95%CI 64–94%) for patients with cRoPE score ≥ 8 (Table 3).

Discussion

This study provides a score counting on demographic and clinical features only, able to accurately predict the presence of a PFO in patients with acute ischemic stroke. The cRoPE score has reliability comparable with the original RoPE score but is independent from neuroradiological findings that limit the application of the latter [4]. Hence, the cRoPE might represent a practical tool already in the acute setting, to help in diagnostic workflow, and even in the place of the original RoPE, when issues with neuroimaging arise. Indeed, neuroimaging fails to detect ischemic lesions in up to 21% of cases [6], rendering the localization of the lesion elusive for scoring purposes [4]. Since cRoPE has the same accuracy and reliability of the original RoPE score, in such clinical setting, it might provide critical insights for the stratification of patients with PFO. Moreover, it could also inform on etiology, with scores 8–10 associated with an 83% attributable fraction, therefore potentially guiding treatment choice.

In addition, cRoPE might also be implemented in transient ischemic attacks, where brain imaging often fails to show hyperacute parenchymal lesions [21].

Our results also underline that cRoPE has high accuracy in predicting the presence of PFO among patients with ESUS. Indeed, while RoPE was developed among patients with CS [4], evidences on the translational potential of such scores to CS subgroups is still lacking, especially among ESUS, a construct which is still highly debated [22]. In this study, cRoPE was 77% accurate in predicting PFO in patients with ESUS, with scores higher than 5 being 73% sensitive and 75% specific in predicting PFO. Therefore, in these cases, PFO might indeed be regarded as a significant contributor to stroke etiology, with consequent diagnostic workup and therapeutic management [23]. Moreover, cRoPE might also have the potential to be implemented as a screening tool to randomize patients to different medical treatment, since the optimal strategy to prevent PFO-related stroke recurrence is still debated, with trials

on ESUS and CS patients possibly flawed by several etiologies mixing up in a miscellaneous category [24]. At the same time, the fact that the accuracy of cRoPE in PFO prediction abruptly decreases considering etiologies other than CS might point toward the need to preserve such instrument for specific circumstances. Indeed, PFO might be regarded as an “innocent bystander” if other etiologies have been unveiled, while it deserves diagnostics when etiology of stroke remains unclear or controversial.

Limitations of this study are represented by the observational design and the relatively small sample size compared with the one proposed for the original RoPE validation. However, the cohort was highly homogeneous among centers enrolling patients, and all patients underwent systematic assessment as well as OCSF and TOAST grouping, allowing for reproducibility and consistency of the data derived from this population. Moreover, the fact that the overall accuracy of cRoPE fits to the one demonstrated for the original RoPE score among patients with CS strongly suggests that our cohort is representative of the general population, and that cRoPE is worth considering in cases of CS and ESUS.

Data availability statement Data sharing request will be considered by the corresponding author.

Compliance with ethical standards

Conflict of interest None.

Ethical approval Protocol was approved by Ethical Committee of Coordinator Centre (4.2011.22), informed consent was obtained, and the study followed standards of the 1964 Declaration of Helsinki.

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